Radiomics Research Study Report

Research Team

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Executive Summary

Hypothesis Status: REJECTED

This radiomics research study aimed to validate whether quantitative imaging features extracted from medical images can reliably distinguish between neuroblastoma and healthy tissue samples. A comprehensive machine learning pipeline was executed, extracting 10 key radiomic features including texture, shape, and first-order statistical characteristics. The modeling phase employed multiple classification algorithms with 10-fold cross-validation, achieving an average accuracy of 60% ± 20% across validation folds. While the model demonstrated moderate predictive capability with some successful classifications (e.g., 87% confidence on true negative predictions), overall performance metrics (AUC: 40% ± 20%, F1-score: 47% ± 27%) indicate limited discrimination power for this classification task with the current feature set and dataset.

1 Study Overview

This investigation examined the efficacy of radiomic features in discriminating neuroblastoma from healthy tissue using machine learning classification.

2 Methodology

2.1 Feature Extraction

10 quantitative radiomic features covering texture, shape, and first-order statistics were extracted from medical images.

2.2 Modeling Approach

Multiple classification algorithms with 10-fold stratified cross-validation were employed to ensure robust evaluation.

2.3 Evaluation Metrics

Comprehensive assessment including accuracy, AUC, recall, precision, and F1-score was performed.

3 Key Results

Table 1: Model Performance Metrics

Metric	Mean Performance	Standard Deviation
Accuracy	60%	±20%
AUC	40%	±20%
F1-Score	47%	±27%

4 Performance Analysis

The model demonstrated variable performance across validation folds, with individual predictions showing confidence scores ranging from 56% to 87%. While some correct classifications were achieved, overall discrimination capability was insufficient for reliable hypothesis validation.

5 Conclusion

Based on the empirical evidence and established performance thresholds for diagnostic classification, the hypothesis that radiomic features can reliably distinguish neuroblastoma from healthy tissue is **rejected**. Further feature engineering, larger datasets, or alternative modeling approaches may be required to achieve clinically meaningful discrimination.

6 Hypothesis Validation Decision

- **Decision:** REJECTED
- **Justification:** Model performance metrics fall below acceptable thresholds for clinical or research validation, with AUC < 0.7 and inconsistent cross-validation results.